

The representation of *s*C clusters: A comparative approach

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Introduction*

Clusters of the shape *s*+consonant (*s*C), exemplified by *stack*, have posed a challenge for theories of syllabification as they defy many of the constraints holding of true branching onsets, as in, for example, *track*. Accordingly, some researchers have proposed that *s* is organized outside the onset constituent that contains the following consonant. Others have proposed that this sort of analysis holds only for a subset of *s*C clusters; those that rise in sonority, for example *slack*, are represented in the same fashion as branching onsets. Yet others have argued that some *s*C clusters, those of the shape *s*+stop, form complex segments.

This chapter will critique each of these proposals. An element shared by all of them is that phonological units are highly articulated: the burden of explanation is placed precisely on the structural relationships that adjacent segments enter into. Although this approach captures many peculiarities of *s*C clusters, there is little attempt to explain why the consonant that displays unorthodox behaviour is typically *s*. Under the view that segments are ordered to maximize their perceptibility, the behaviour of *s* becomes less puzzling: strident fricatives have robust internal cues, ensuring their perceptibility even in non-optimal contexts.

If the acoustic properties of *s* are of central importance, it behooves us to ask whether the differences between *s*C clusters and branching onsets can be explained solely by perceptual considerations. This, of course, would challenge the view that a structural approach to cluster well-formedness is necessary. In the final section of the chapter, I will argue that this position is too strong. I will conclude that an adequate understanding of *s*C clusters requires consideration of both perceptual and structural factors.

Much of the chapter compares *s*C clusters with branching onsets. We will observe that they differ in several respects: phonotactic constraints, word-internal syllabification, allomorph selection, patterns of reduplication, options for cluster repair, etc. Although the general observations we will detail are likely to be accepted by most phonologists, there is little agreement on how these differences should be formally represented. In this context, there are three topics that will be addressed.

The first involves critical assessment of various proposals in the literature concerning the representation of *s*C clusters –as a single class– in contrast to obstruent+sonorant clusters. Henceforth, I will use the term ‘obstruent’ to refer to obstruents other than *s*. ‘*s*’ is itself a cover term for the sibilant(s) appearing in *s*C clusters; although this sibilant is usually /s/, in some languages, other sibilants pattern as *s* (e.g., German *s* is usually /ʃ/; in Russian, /s, z, ʃ, ʒ/ all pattern as *s*).

Concerning obstruent+sonorant clusters, there will be nothing particularly special to say about their representation; they form branching onsets and there is little controversy on this matter among those who accept a hierarchically-organized syllable. For *s*C clusters, in contrast, several options will be considered. Most of these share the idea that *s* is an appendix, a segment that is not organized by any sub-syllabic constituent; one views *s* as a coda. We will see, in addition, that some researchers propose a single representation for *s*C in all languages; others argue for different representations across languages.

The second topic that must be addressed is whether, in a given language, all *s*C clusters are represented in the same fashion. *s*+sonorant clusters are phonotactically ambiguous: like obstruent+sonorant clusters, they rise in sonority; yet, like *s*+obstruent clusters, they do not respect

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the place constraints holding of obstruent+sonorant clusters. Depending on the weight assigned to each of these, different conclusions will be arrived at concerning the analysis of *s*+sonorant. For those researchers who place most weight on sonority profile, *s*+sonorant clusters form branching onsets. This research itself falls into two categories. One body of work aims to show that *s*+sonorant patterns with branching onsets while *s*+obstruent patterns differently and is organized with some type of appendix. Another body of work considers *s*+sonorant clusters to be branching onsets, but focuses on arguing that *s*+stop clusters form complex segments.

The proposals sketched above assume that the syllable is hierarchically organized. However, there is a growing literature which de-emphasizes the role of constituency and aims to provide phonetically-grounded explanations for phonological behaviour. The third topic therefore considers whether differences in the behaviour of obstruent+sonorant, *s*+sonorant and *s*+obstruent can be explained by perceptual considerations alone. This topic will be the focus of the final section of the chapter. Until then, a structural approach will be assumed.

Cluster phonotactics

We begin by detailing the phonotactic constraints most commonly held of obstruent+sonorant clusters on the place and sonority dimensions, in turn, examining *s*C clusters on these same dimensions (see also Chapter wbctp0033; Chapter wbctp0055). Our focus will be on the left word edge; other types of phonological behaviour will be discussed in later sections when we examine alternative representations for *s*C clusters.

Consider the inventories of two-member clusters found in word-initial position in Dutch in (1).¹ The data are organized by the place and manner values of *C*₁ for obstruent+sonorant clusters and of *C*₂ for *s*C clusters. (On cluster phonotactics for Dutch, see Trommelen 1984, van der Hulst 1984, Fikkert 1994, Booij 1995, van der Torre 2003.)

(1) Dutch:

a. Obstruent+sonorant:

	*tn	kn
pl	*tl	kl
pr	tr	kr
fl		χl
fr		χr

b. *s*C clusters:

sp	st	
		sχ
sm	sn	
	sl	
	*sr	

We first consider place identity which forbids the consonants in obstruent+sonorant clusters from having the same place (Chapter wbctp0022). This captures the ill-formedness of */tn, tl/ in

¹ We restrict discussion of obstruents in clusters to those that are voiceless; some languages display fewer options for voiced obstruents (e.g., */vl, vr/ vs. /fl, fr/ in English). We avoid consonant+glide clusters altogether as there are more representational options available for glides than we have space to consider.

(1a).² Turning to (1b), the well-formedness of /st, sn, sl/ indicates that *sC* clusters do not respect place identity, suggesting that *sC* clusters do not have the same representation as obstruent+sonorant clusters. However, before we can conclude this with certainty, we must consider */sr/, which is ill-formed in Dutch, regardless of whether /r/ is coronal or dorsal (which differs across varieties; Sebrechts 2015).³ This suggests that the ill-formedness of this cluster has nothing to do with place identity. We return to */sr/ later in the chapter.

A second less commonly discussed constraint on place concerns asymmetries that hold between *C*₁ and *C*₂ in obstruent+sonorant clusters (when *C*₂≠glide). In languages like English, this is not very revealing because *C*₂ is restricted to liquids which are coronal (*tl, *θl). Dutch is (potentially⁴) more illuminating because it contains stop+nasal clusters and nasals contrast for place. As (1a) shows, when a nasal is in *C*₂ in an obstruent+sonorant cluster, it must be coronal: /kn/ is well-formed; */km/ is out. The broader generalization is thus that when *C*₂ is a contoid, it must be coronal (unless /r/; see note 2).

*C*₂ in *sC* clusters has a different profile: it can have any place of articulation. Indeed, in *s*+obstruent and *s*+nasal clusters, *C*₂ displays the same range of place contrasts attested for singleton onsets (e.g., /sp, st, sʎ/ alongside /p, t, k/ in Dutch). Directly comparing nasal-final clusters in Dutch, we arrive at the following: */km/, ✓/kn/; ✓/sm/, ✓/sn/. The absence of */km/ alongside the presence of /sm/ is unexpected if these clusters are represented identically.

Because of the disputed status of Dutch /kn/ (note 4), we turn to Modern Greek to better examine differences in the place profile between *C*₂ in obstruent+sonorant versus *sC* clusters. (On Modern Greek cluster phonotactics, see Joseph & Philippaki-Warbuton 1987, Drachman 1990, Klepousniotou 1998, Morelli 1999, Tzakosta & Vis 2009; on Attic Greek, see Steriade 1982.) The data in (2) reveal that, although *C*₂ in an obstruent+sonorant cluster can have any manner leading to a wider range of cluster profiles than in Dutch, *C*₂ must still be coronal (Klepousniotou 1998).^{5,6} (2b) shows that *sC* clusters are more restricted on the manner dimension than they are in Dutch, but among clusters with obstruents in *C*₂, it can nevertheless be seen that *C*₂ can have any place.

² Place identity is not respected with /r/. In Dutch varieties with coronal /r/, coronal+/r/ clusters are well-formed, as are dorsal+/r/ clusters in varieties with dorsal /r/. Even in languages where /t/ and /r/ are articulated near identically, the constraint is not respected (see Arvaniti 2007 on Greek). This may suggest that /r/ permanently lacks place (Rice 1992, Goad & Rose 2004; see also Chapter wbctp0030).

³ Concerning Dutch */sr/, some speakers realize /sʎr/ as [sr] (Waals 1999). If this represents a reanalysis of /sʎr/ (van der Torre 2003), then /sr/ is well-formed for these speakers.

⁴ I say potentially because, as is undoubtedly evident, obstruent+sonorant clusters will be analysed as branching onsets below. There is, however, dispute about the status of /kn/ in Dutch, as branching onset (Fikkert 1994, Booij 1995) or appendix-initial (Trommelen 1984, van der Hulst 1984, Kager & Zonneveld 1985/86). Notably, intervocalic /kn/ is syllabified as coda+onset, counter to the branching onset analysis.

⁵ Thanks to Jenny Dalalakis and Katerina Klepousniotou for help with the Greek data.

⁶ Stop+stop and fricative+stop are often considered to be archaic in spoken Modern Greek (Joseph & Philippaki-Warbuton 1987, Morelli 1999). They do occur in higher registers which is why they are included here. /ps, ts, ks/ are absent from (2a); I assume that they are complex segments (following Tzakosta & Vis 2009). Cluster in parentheses (as well as /tm/) are not productive, although the number of /sm/-initial roots is somewhat larger than for the others.

(2) Modern Greek:

a. Obstruent+sonorant:

pt		kt
(pn)	*tn	(kn)
pl	*tl	kl
pr	tr	kr
ft	*θt	xt
fθ		(xθ)
*fn	(θn)	xn
fl	(θl)	xl
fr	θr	xr
(mn)		

b. sC clusters:

sp	st	sk
sf	(sθ)	sx
(sm)	(sn)	
	*sl	
	*sr	

In sum, we have observed that C_2 in an obstruent+sonorant cluster does not parallel C_2 in an sC cluster on the place dimension. On the contrary, a closer parallel is observed between C_1 in an obstruent+sonorant cluster and C_2 in an sC cluster, which we return to below.

We consider finally the sonority constraints that hold between C_1 and C_2 in initial clusters.⁷ Greek is not very revealing here, as both obstruent+sonorant and sC clusters can have a falling, flat or rising sonority profile (although (2b) indicates that the productivity of the latter for sC clusters is questionable; we return to this below). We therefore focus on Dutch. (1) shows that sC clusters need not rise in sonority, in contrast to obstruent+sonorant clusters. Although C_1 in an sC cluster is a fricative and so the potential exists for these clusters to be limited to those that rise in sonority, this is not what is observed. Dutch contains both falling sonority s +stop and flat sonority $/s\chi/$.

sC clusters ≠ branching onsets

Our examination of phonotactic constraints has revealed that obstruent+sonorant and sC clusters pattern differently on three dimensions: place identity, C_2 place profile and C_1C_2 sonority profile. Hereafter, we turn to other types of phonological behaviour. Our first goal is to motivate the position that sC clusters are represented differently from obstruent+sonorant clusters. We then delve into the various options for sC clusters. For both of these topics, we will focus on phenomena where sC clusters pattern as a single class. The following section will then examine evidence that s +sonorant and s +obstruent clusters may be organized differently, s +sonorant clusters as branching onsets and/or s +stop as complex segments.

In the preceding section, we observed that obstruent+sonorant clusters have a profile consistent with the prototypical syllable, one that rises in sonority toward the peak. This indicates that clusters

⁷ I assume the following sonority scale, which is roughly based on relative intensity: stop < fricative < nasal < liquid < vocoid. For critical discussion of sonority, see the contributions to Parker (2012) and Ball & Müller (2016); see also Chapter wbctp0049.

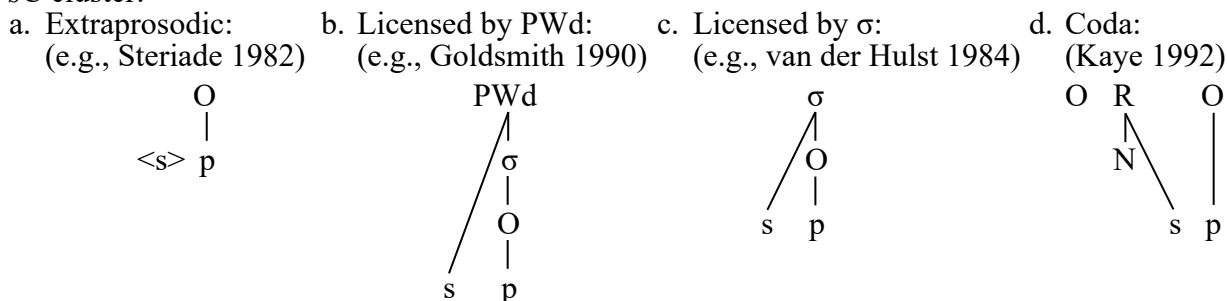
of this shape (modulo language-specific constraints on place identity and sonority distance) are organized as branching onsets; see (3).

(3) Branching onset:



Following Kaye, Lowenstamm & Vergnaud (1990), I assume that onsets are universally left-headed. Heads can host a range of segmental contrasts; dependents are segmentally restricted. Considering place contrasts, earlier discussion of C_2 place profile revealed that there is more in common between C_1 in a branching onset and C_2 in an sC cluster. This suggests that sC clusters are right-headed. If onsets are left-headed, this de facto places s outside of this constituent, as reflected in the representations in (4). In (4), skeletal slots have been suppressed to facilitate comparison across theories; PWd abbreviates Prosodic/Phonological Word (see Chapter wbctp0051).

(4) sC cluster:



In (4a), s is extraprosodic. I am using this term in its narrowest sense, to refer only to the situation where an element is licensed but not organized into higher structure; compare (4b-c). The representations in (4b-c) share with (4a) the idea that s is an appendix: s is not organized by any sub-syllabic constituent, in contrast to (4d) where s is a coda (technically, in Kaye's framework of Government Phonology (GP), a rhymal dependent).

Two predictions follow from the difference in representation and headedness in (3) versus (4). One, there should be languages that permit branching onsets but not sC clusters and vice versa. This prediction holds true. Spanish and Basque are languages with branching onsets but lacking initial sC clusters (Harris 1983, Hualde 1991). Itonama (isolate; NE Bolivia) and Misantra Totonac (Mesoamerican; Veracruz, Mexico) have the opposite profile: initial clusters are restricted to sC (Liccardi & Grimes 1968, MacKay 1994). Relatedly, Fikkert's (1994) study on the acquisition of Dutch reveals that some children acquire branching onsets first, parallel to Spanish and Basque, while others acquire sC clusters first, parallel to Itonama and Misantra Totonac. Two, languages permitting both types of structures should not prevent them from being combined. To my knowledge, this prediction always holds. In languages that have branching onsets and sC clusters, three-member clusters of the shape s +branching onset are also well-formed. However, these clusters may be restricted to a subset of what would be expected from a free combination of sC clusters and branching onsets in the particular language (e.g., Greek: ✓/sx/, ✓/xr/, */sxr/; English: ✓/sk/, ✓/kl/, */skl/ (loans aside)).

Further, an explanation emerges under (4) for why the constraints against place identity and for rising sonority do not hold of sC clusters. It is not enough for two consonants to be adjacent; they must be sisters, as in (3).

Finally, we demonstrate that the difference in headedness between branching onsets and sC clusters can account for a commonly-attested pattern of cluster reduction in acquisition illustrated

in (5) from two learners of German and English respectively, Annalena (Elsen 1991) and Amahl (Smith 1973).⁸

(5)	Annalena (age 1;4-1;9)			Amahl (age 2;2-2;6)	
	Output	Target	Gloss	Output	Gloss
obs+son	[daʊbɐ]	[tr]aube	‘grape’	[bɛ:t]	‘plate’
	[fikə]	[fl]iege	‘fly’	[gi:m]	‘cream’
s+obs	[piɡəl]	[ʃp]iegel	‘mirror’	[baidə]	‘spider’
	[damə]	[ft]ein	‘stone’	[ɡip]	‘skipping’
s+son	[mɪsən]	[ʃm]eißen	‘to throw’	[ni:d]	‘sneezed’
	[la:fə]	[fl]afen	‘to sleep’	[lʌg]	‘slug’

In (5), the head of the cluster survives, regardless of its position in the string or its relative sonority. This suggests that some learners recognize that branching onsets are left-headed and that *s*C clusters are de facto right-headed, even though initial clusters are absent altogether from their outputs (Goad & Rose 2004).

Alternative representations for *s*C clusters

Thus far, we have motivated different representations for branching onsets and *s*C clusters, based on evidence from phonotactics and cluster reduction in acquisition. We turn now to critique the various options for *s*C clusters in (4).

In (4a), *s* is extraprosodic.⁹ Although it is unaffiliated to any syllable constituent (at least at early levels of the phonology), it is licensed and thus not subject to stray erasure. This structure is motivated by, for example, Steriade (1982) with data from Attic Greek, Sanskrit, Latin and English; Vennemann (1982) and Wiese (1988) from German; Kager & Zonneveld (1985/86) from Dutch; Chierchia (1986) and Davis (1990) from Italian.

Although (4a) captures the unique behaviour of *s*C clusters, it cannot easily express the observation that some languages have different types of extraprosodic segments. In English, for example, the same status should not be assigned to /s/ in *s*C clusters as to inflectional /s/ in (6a), yet both fall outside of core syllabification. Further, on the face of it, some languages permit adjacent extraprosodic segments. English allows an extra morpheme-internal position at the right edge of words that is not permitted word-internally which some researchers have analysed as extrarhymal (e.g., Myers 1987). Since such words can also be inflected, (6b), the result would appear to be two adjacent extraprosodic positions (see also Chapter wbctp0036).¹⁰

- (6) English:
- a. <s>træp<s> ‘straps’
 - b. tæk<s><t> ‘taxed’
 - træm<p><s> ‘tramps’

If the final two positions in (6b) are both licensed as extraprosodic at the same point, the result would violate the Peripherality Condition: extraprosodic status can only be assigned to elements in peripheral positions (Hayes 1981, Harris 1983). The alternative that is typically adopted,

⁸ [b̥ d̥ ɡ̊] indicate voiceless unaspirated lenis stops in Amahl’s data (Smith 1973). Overdots are used by Elsen (1991) to indicate ambisyllabic consonants in Annalena’s data.

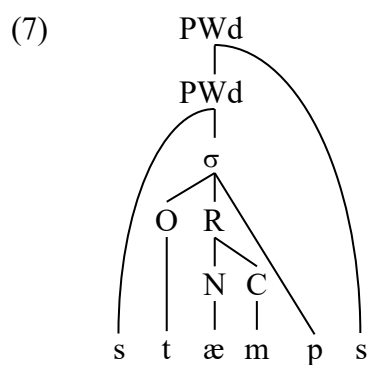
⁹ (4a) disregards the fact that scholars working with this type of representation hold different views on whether the syllable is internally structured.

¹⁰ More troubling are words like [nɛkst] ‘next’ with two morpheme-internal extraprosodic positions; we return to such cases when we examine the possibility that *s*+stop form complex segments.

therefore, is that these two types of extraprosodic segments are extraprosodic at different points in the derivation. Extraprosodic /s/ in ‘taxed’ and /p/ in ‘tramps’ are incorporated as codas in the postlexical phonology before inflectional /t/ and /s/ are added (cf. Borowsky 1986). Initial /s/ in (6a) similarly loses extraprosodic status in the postlexical phonology where lexical constraints on sonority profile are no longer assumed to hold. The problem with this approach, however, is that it fails to show that extraprosodic elements ever function as members of the constituents that ultimately come to organize them, the onset in (6a) and the coda in (6b) (Piggott 1991).

This problem is resolved once the Strict Layer Hypothesis is abandoned, the requirement that elements be dominated by the immediately higher category in the prosodic hierarchy (Nespor & Vogel 2007). Extraprosodic segments on this view are technically not *extraprosodic* (unaffiliated); instead, they are organized by some higher constituent in the prosodic hierarchy.

One possible representation for <s>tæm<p><s>, consistent with this approach, is in (7): initial /s/ is organized by the PWd, extrarhymal /p/ is organized by the syllable and inflectional /s/ is adjoined to the PWd. This representation is consistent with the Peripherality Condition: extrarhymal /p/ and inflectional /s/ are each at the right edge of a separate PWd (Goad & White 2006).



Returning specifically to *sC* clusters, the representation for /st/ in (7) involves *s* linking directly to the PWd (see, e.g., Goldsmith 1990 drawing on evidence from English; Trommelen 1984 and Fikkert 1994 from Dutch; Goad 2012 from Italian). An alternative involves *s* linking directly to the syllable, the inverse of extrarhymal /p/ in (7) (see, e.g., van der Hulst 1984 with evidence from Dutch; Levin 1985 and Kenstowicz 1994 from English; Barlow 1997 and Gierut 1999 from English in phonologically-delayed children; Drachman 1989 and Tzakosta & Vis 2009 from Greek; Tzakosta 2009 from child Greek)).¹¹ These two proposals were provided earlier as (4b) and (4c), respectively.¹²

The alternatives in (4b-c) make different predictions concerning the distribution of *sC* clusters. I show that both options (or their equivalents) are needed, indicating that *sC* clusters cannot be represented identically across languages (Goad & Rose 2004, Ewen & Botma 2009, Vaux & Wolfe 2009). In languages with (4b), *sC* is only licensed PWd-initially. Italian has this profile (Goad 2012). *sC* clusters can only occur at the left edge of the word or at the left edge of the stem in prefixed forms, as shown in (8). If prefixes are adjoined to the PWd (Peperkamp 1997), then this restriction on *sC* distribution can be captured with (4b). We will see shortly that, in word-medial position, *s* in *sC* clusters is always parsed as a regular coda in Italian, which will be shown to follow from the representation for initial *sC* in (4b).

¹¹ Note that Levin (1985) adjoins, rather than directly links, *s* to the syllable. To the list in the text we can add Giegerich (1992), Hall (1992) and Booij (1995) who analyse *s* as an onset-internal appendix, using data from English, German and Dutch respectively; and Ewen & Botma (2009) who organize *s* into the specifier position of the onset for Germanic.

¹² Other less commonly proposed licensers for *s* will not be considered due to space constraints: the Foot (Green 2003 on Munster Irish) and the Phonological Phrase (Vaux & Wolfe 2009 on Armenian).

- (8) PWd-initial *s*C in Italian:
 [stá:to]_{PWd} ‘state, condition’
 [kon[statá:re]_{PWd}]_{PWd} ‘to certify’

In contrast to Italian, *s*C clusters in Dutch and English have a wider distribution in word-medial contexts: in addition to words with regular coda *s*, both languages contain monomorphemic words where the rhyme preceding *s*C appears to be “full” and, thus, unable to accommodate *s* as a coda (e.g., Dutch [ékstər] ‘magpie’, English [ékstrə] ‘extra’ (van der Hulst 1984, Levin 1985)). We will see that this is evidence that *s*C requires the representation in (4c) for these languages.

We focus on English, examining word-medial clusters in some detail. Harris (1994) documents the constraints governing three-position rhymes shaped VVC in this language. As (9a) reveals, coda sonorants in these superheavy syllables are confined to coronals that share place with the following onset (*[ʃoʊlbər], *[maʊmpən]). PWd-internal rhymes shaped VCC are not considered by Harris (they are not well-formed in GP, the framework in which he works). (9b) shows that the onset is similarly constrained to coronal and the preceding consonants must be homorganic nasal+stop (*[vɪltnər], *[dʒʌŋ(k)ʃən]).

- (9) English:
- | | |
|----------------------|-------------------------|
| a. VVC rhymes: | b. VCC rhymes: |
| [ʃoʊldər] ‘shoulder’ | [æntlər] ‘antler’ |
| [maʊntən] ‘mountain’ | [vɪntnər] ‘vintner’ |
| [kaʊnsəl] ‘council’ | [dʒʌŋ(k)ʃən] ‘junction’ |

With these constraints in mind, we turn to cases where the consonant following VV/VC is *s*. Parallel to (9a), (10a) shows that only coronal *s* is permitted after VV (*[i:ftər]) and the following consonant must be coronal (*[i:spər]).¹³ This leads Harris to conclude that *s* in (10a) is syllabified as the coda of a three-position rhyme. The forms in (10b) are similarly parallel to those in (9b), seemingly leading to the same conclusion.

- (10) a. VVs rhymes:
- | | |
|-------------------|-----------------------|
| [i:stər] ‘Easter’ | b. VCs rhymes: |
| [ɔ:stər] ‘oyster’ | [manstər] ‘monster’ |
| | [mɪnstrəl] ‘minstrel’ |

Problems arise, however, in (11). (11a) show that VCs does not always respect the constraints holding of (10b): the consonant preceding *s* is not restricted to place-sharing sonorants. (11b) reveals that the onset following *s* can be other than coronal, in contrast to (10a-b). One could object on grounds that, *extra* aside, the words in (11) involve Latinate prefixes. However, these prefixes are not synchronically productive: they fall within the stress domain and must therefore be contained inside the lower PWd ([ábstəkəl]_{PWd}, *[áb[stəkəl]_{PWd}]_{PWd}).

- (11) Appendix *s*:
- | | |
|---------------------------|--|
| a. Non-coronal codas: | |
| [ékstrə] ‘extra’ | |
| [ábstəkəl] ‘obstacle’ | |
| b. Non-coronal onsets: | |
| [èkspəzɪʃən] ‘exposition’ | |
| [kánskɹɪpt] ‘conscript’ | |

If the forms in (11) truly involve appendix *s*, we expect to find *s* occurring after three-position rhymes of the shape in (9). (12) shows that such words are well-formed, albeit rare. *bolster* and

¹³ Neither constraint holds in dialects with lengthened [a:]/[æ:]: *a:fter*, *ba:sket* (Harris 1994).

holster are monomorphemic, and while *-ster* in *upholster* is historically a class 2 suffix (C17: *uphold-ster* ‘small furniture dealer’), this analysis no longer holds, as revealed by the fact that *upholster* is now a verb and *-y* can attach outside (*upholstery*).¹⁴

- (12) VVC rhymes followed by *s*:
- | | |
|----------------|-------------|
| [boulstər] | ‘bolster’ |
| [houlstər] | ‘holster’ |
| [ʌp(h)oulstər] | ‘upholster’ |

In short, while some instances of medial *sC* in English may involve *s* as coda, appendix *s* is required to capture the data in (11) and (12). This thereby supports the postulation of (4c) where *s* is licensed by the syllable.

Although languages like English appear to require (4c), this analysis cannot straightforwardly capture the fact that these same languages syllabify *s* as a coda after short stressed vowels. If the appendix representation is available word-medially, why do native speakers judge *sC* as heterosyllabic (*pés.ter*) rather than as appendix+onset (*pé.ster*)? It cannot be due to sonority profile, as will be seen shortly for Dutch. Heterosyllabicity of medial *sC* is handled more elegantly in Kaye’s (1992) GP approach to *sC* clusters which we consider now.

Kaye (1992) proposes that *sC* clusters are syllabified as coda+onset sequences, shown earlier in (4d).¹⁵ He provides support from Italian, Ancient Greek, European Portuguese and British English; see further Brockhaus (1999) on German and Cyran & Gussmann (1999) on Polish. We first consider Italian where the coda+onset pattern observed for *péster*-type words is illustrated more concretely. In Italian, rhymes of stressed syllables must branch (Chierchia 1986). When a stressed syllable lacks a coda, the vowel lengthens; see (13a-b). (13c) shows that *sC* clusters do not trigger lengthening as do branching onsets; instead, they pattern with coda+onset sequences, (13d), revealing that medial *s* is a coda.

- (13) Medial *sC* in Italian:
- | | | |
|----|----------|---------|
| a. | [fá:to] | ‘fate’ |
| b. | [ká:pra] | ‘goat’ |
| c. | [pásta] | ‘pasta’ |
| d. | [párko] | ‘park’ |

Turning to word-initial *sC*, Kaye proposes that *s* in this position is also a coda; the difference between the initial and medial environments is that, in the former case, *s* is the coda of an empty-headed syllable. The coda analysis for word-initial *s* follows from the Uniformity Principle in GP, which requires syllabification to be constant for a given string of segments, within and across languages. Kaye provides empirical support for Uniformity from Italian masculine definite article allomorphy and *raddoppiamento sintattico*. In the former, vowel- and *sC*-initial words pattern together, in contrast to words beginning with (branching) onsets; compare (14a-b) with (14c).

¹⁴ Hayes (2009:210-211) considers the vowel in such words to be monophthongized (so presumably short) in some dialects. However, this does not hold of all dialects; witness, for example, RP [bəʊlstər].

¹⁵ Coda *s* must be preceded by an empty nucleus to constitute a well-formed syllable. A precursor to this can be found in Vennemann (1988), who proposes that initial *s* in *sC* clusters is “quasi-nuclear”, a type of degenerate syllable. He argues that this analysis of Latin *s* explains its development into a regular *Vs* syllable in some Romance languages like Spanish. Working within Strict CV (Lowenstamm 1996), Polgárdi (2022) takes a similar approach for *sC* clusters in Italian, Portuguese and English: *s* under C branches onto an immediately adjacent V slot and so is effectively quasi-nuclear. *s* in *sC* and *Cs* clusters is argued to be fully nuclear in Blackfoot by Goad & Shimada (2014), building on ideas in Denzer-King (2009 and Derrick (2006); see also Weber (2020).

Since the representation for *s* in an *sC* cluster in (4d) includes a preceding nucleus, there is a structural parallel with vowel-initial words.

(14) Masculine definite article allomorphy (Davis 1990):

- a. l'est /lo est/ 'the east'
- b. lo studente 'the student'
- c. il burro 'the butter'
- il clima 'the climate'

In *raddoppiamento sintattico*, the first consonant in an onset geminates when the preceding word ends in a stressed vowel, while the first consonant in an *sC* cluster resists gemination; see (15). The pattern in (15b) follows directly from the view that *s* is a coda: as coda+onset, *sC* already has precisely the structure that holds of geminates.

(15) Raddoppiamento sintattico (Chierchia 1986):

- a. paltó pulíto [paltóppulító] 'clean coat'
- cittá trístete [ʧittáttrístete] 'sad city'
- b. cittá straniéra [ʧittásttraniéra], *[ʧittásstraniéra] 'foreign city'

Although it is evident from (14) and (15) that *sC* clusters in Italian cannot be analysed in the same fashion as branching onsets, the proposal that they contain an initial appendix can also handle these data (see Chierchia 1986, Davis 1990). Of the cases that Kaye considers, the construction that poses a particular challenge for appendix *s* is European Portuguese vowel nasalization (Goad 2012). We turn to this case now.

In European Portuguese, nasal consonants cannot close syllables. While /n/ is realized intact before vowel-initial bases, (16a), before onset-initial bases, nasality surfaces on the preceding vowel, (16b-c). Interestingly, *sC*-initial bases pattern as vowel-initial, (16d).

(16) European Portuguese:

- a. [in]admissível 'inadmissible'
- b. [ĩ]pureza 'impurity'
- [ĩ]satisfeito 'dissatisfied'
- c. [ĩ]tratável 'unsociable'
- d. [in][k]apável 'inescapable'

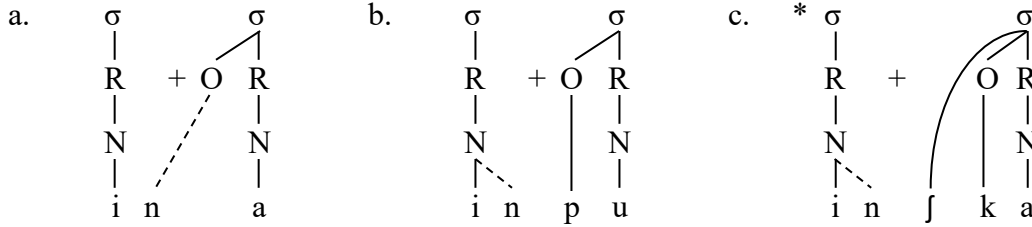
(16d) can be straightforwardly expressed under Kaye's view that *sC* clusters are coda+onset because, for independent reasons, all syllables in GP contain an onset constituent. Consider the representations below. In (17a), /n/ associates to the onset of the first syllable in the base. In (17b), this position is occupied, so nasality is preserved on the preceding vowel. The right result obtains in (17c) precisely because the syllable containing *s* includes an empty onset.

(17) Coda analysis:

- a. O R + O R b. O R + O R c. O R + O R O R
- | | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| N | N | N | N | N | N | N | N |
| | | | | | | | |
| i | n | a | i | n | p | u | f |
- i n a i n p u i n f k a

An appendix analysis of *s*, it seems, cannot formally capture (16d). See (18c) where bases with initial *sC* are incorrectly predicted to pattern with onset-initial bases because there is no empty constituent to host /n/.

(18) Appendix analysis:



Kaye's paper compares the coda+onset analysis of *sC* clusters to the alternative that they form branching onsets; the option that *s* is analysed as an appendix is not discussed. We have seen that appendix *s* cannot straightforwardly capture European Portuguese. It also goes against the Uniformity Principle: Italian *s* is an appendix in *straniéra* but a coda in *pásta*; this would likely be considered a weakness by proponents of GP. However, there are contexts where word-internal *s* maintains its appendix status, in contrast to the pattern observed for Italian; *sC* can follow a rhyme that is already full, so *s* cannot be accommodated as an ordinary coda. We have already observed this for English, but we have also seen that English permits word-internal three-position rhymes under limited circumstances and most of the problematic *sC* data are in words that historically involve prefixes. To ensure that there is nothing unusual about English concerning the distribution of *sC* clusters, let us turn to Acoma (Miller 1965).

Initial *sC* clusters in Acoma are shown in (19a). (Note that *s* has three allophones: [s], [ʃ], [f].) Two-position rhymes in Acoma are limited to VV and seemingly Vs (loans aside); see (19a-b). If word-internal *s* were always a regular coda, as it appears to be in (19b), we would expect it to be restricted to occurring after short vowels, like in Italian. (19c) shows, however, that this is not the case.

(19) Acoma:

- a. [ʃpúuná] 'pottery'
[ʃfáits^{hi}i] 'it is muddy'
- b. [suft'á] 'I took water'
[ʔéʃká] 'rawhide'
- c. [ʔúuʃfúuts^{hi}i] 'drum'
[kúuʃf^haʃa] 'knot'

Taken together, the data in (19) suggest that *s* is an appendix to the syllable in Acoma, rather than a regular coda. Before we accept this analysis, though, an alternative must be considered, that *sC* clusters are bogus clusters in Acoma, that they are adjacent onsets interrupted by an empty nucleus. This analysis would allow *s* to occur after long vowels and it could be motivated by the observation that the plain-aspirated contrast is maintained after *s* in Acoma (Goad 2012).¹⁶ Productions like the latter are unusual in *sC* clusters, as the spread glottis present in *s* will have narrowed by the time the stop closure is released, such that no aspiration will be observed on the stop (Iverson & Salmons 1995). Evidence from the right word edge, however, reveals a potential challenge: word-final consonants, which are always onsets of empty-headed syllables in GP (Kaye 1990), are not permitted in Acoma. If *s* in medial *sC* strings is represented as *s*Ø, the absence of word-final *s*, which would also be analysed as *s*Ø, is surprising.

¹⁶ The same representation may hold for English [sf] in *sphere* vs. [sv] in *svelte*, where the voicing contrast is surprisingly maintained after *s*. Because *s*+fricative exists only in a handful of loanwords in English, the regular representation required for initial *sC* may not be available for this string and the language may instead opt for *s*ØC.

In sum, we have critiqued four representations for *sC* clusters. We have also considered a fifth option, that *sC* is not actually a cluster, but instead, an empty position intervenes between *s* and *C*. Amongst the four true cluster options for *sC*, we can likely dispense with the extraprosodic analysis for *s* in (4a) in favour of the alternatives in (4b-d), but selecting between these alternatives is no easy task. On one hand, there are languages like English and possibly Acoma that are most compatible with the position that *s* is an appendix to the syllable; on the other are languages like European Portuguese and to a lesser extent Italian that seem to require a coda analysis for *s*. When Italian is compared with English and Dutch, it becomes clear that languages also differ in where *sC* can occur, stem-initially only or also morpheme-internally, revealing that appendix *s* must be licensed by the PWd in the former case and by the syllable in the latter.

Are all *sC* clusters represented in the same manner in a given language?

Thus far, we have treated all *sC* clusters as a single class. Given that these clusters fall into two groups concerning their sonority profile, several researchers have proposed different representations for each group within a given language. One body of work considers rising sonority *sC* clusters to be structured as branching onsets while *s*+obstruent clusters are appendix-initial (see, e.g., Hall 1992 with data from German; Fikkert 1994, Booij 1995 from Dutch; Gierut 1999 from English in phonologically-delayed children). Another body of work motivates a different analysis for *s*+stop clusters: they are not actually clusters but, instead, form complex segments (see, e.g., van de Weijer 1996 drawing on data from several languages; Fudge 1969, Fujimura & Lovins 1982, Ewen 1982, Selkirk 1982 from English; Broselow 1983 from L2 English; Wiese 1996 from German; Barlow 1997 from English in phonologically-delayed children). Since all of this research draws a boundary between rising sonority clusters and *s*+stop clusters, much of the data effectively shows evidence for either position.

The section on cluster phonotactics revealed that, when place of articulation is examined, end-state grammars challenge the view that *s*+sonorant clusters form branching onsets. Evidence from acquisition (Chapter wbctp0101), however, suggests that some learners disregard place, relying solely on sonority in their analysis of left-edge clusters: all rising sonority clusters –both obstruent+sonorant and *s*+sonorant– pattern together. Consider the Dutch children in Fikkert’s (1994) study: all follow the same developmental path for fricative+liquid and *s*+liquid clusters. The stages through which learners pass are exemplified in (20) with data from Jarmo.

(20) Jarmo’s cluster development: Fricative+liquid and *s*+liquid targets:

Stage	Consonant retained	Examples			Age
1	fricative/ <i>s</i> (as stop)	/ˈvlɪndər/	[ˈkɪnʊ]	‘butterfly’	2;2.6
		/ˈsla:pə(n)/	[ˈta:pə]	‘to sleep’	2;0.28
2	liquid	/ˈvlɪndər/	[ˈlɪŋʊ]	‘butterfly’	2;2.27
		/ˈsla:pə(n)/	[ˈla:pə]	‘to sleep’	2;3.9
3	fricative/ <i>s</i>	/ˈχlɛiˌbaːn/	[ˈχɛiχaːn]	‘slide’	2;4.1
		/ˈsla:pə(n)/	[ˈsa:pə]	‘to sleep’	2;3.9
4	fricative/ <i>s</i> +liquid	/ˈflɛsjə/	[ˈslɛsjə]	‘bottle’ (dim)	2;4.1
		/slak/	[flak]~[slak]	‘snail’	2;4.1

Further, all children in Fikkert’s study master *s*+stop clusters at a different point in time from rising sonority clusters. As all rising sonority clusters pattern together in Dutch acquisition, in contrast to *s*+obstruent, Fikkert concludes that *s*+sonorant clusters are represented in the same manner as branching onsets. For her, *s*+obstruent clusters involve *s* licensed by the PWd, (4b) above.

While sonority plays a decisive role in Fikkert’s data, this is not the case for all children. Indeed, we observed in (5) that when only one member of a cluster is produced by Annalena and Amahl, it is the cluster head that survives, regardless of its relative sonority. To show that this pattern extends past the deletion stage, consider the developmental path for Amahl in (21).

(21) Amahl's cluster development:

Stage (Age)	obstruent+liquid	/sl/	/sm sn/	/sp sk/	/st/
1-8 (2.60-2.175)	reduction to head	reduction to head	reduction to head	reduction to head	reduction to head
13-14 (2.233-2.256)	branching onset acquired				
15-19 (2.261-2.333)		fusion	fusion	vacuous fusion	vacuous fusion
20-22 (2.345-3.38)					fusion
24 (3.78-3.96)		appendix acquired			
25 (3.104-3.128)			appendix acquired		
26-29 (3.133-3.355)				appendix acquired	appendix acquired

All clusters reduce to the head through stage 8. Branching onsets are acquired first, emerging at stage 9 and being fully mastered at stage 13. *sC* clusters are reduced to the head until stage 15 at which point the two consonants undergo fusion. Fusion is overt for *s*+sonorant (e.g., /sn/→[n̥]), but not initially for *s*+stop since both consonants are voiceless (e.g., /st/→[t]). At stage 20, when stridency is acquired, fusion becomes overt for /st/ (/st/→[t^s~s~ɬ]). Importantly, at this stage, no target *s*+sonorant clusters are realized as [s]; thus, earlier reduction to the head cannot have been due to the unavailability of [s] in Amahl's productions. Appendices are not acquired until 210 days later than branching onsets, first for /sl/. The remaining *sC* clusters are then acquired over a period of 55 days.

The developmental profiles for Jarmo and Amahl are completely different for *s*+sonorant clusters. It truly appears that these clusters are analysed as branching onsets in Jarmo's grammar and with an initial appendix/coda in Amahl's grammar. We briefly consider the consequences of this.

In the section on phonotactics, we observed that place identity and C₂ place profile suggests that, in end-state Dutch, all *sC* clusters are represented differently from branching onsets. If learners like Jarmo initially only consider sonority in assigning clusters to categories, then some unlearning must take place to arrive at the target representations. These children must discover that: (i) place identity is respected in branching onsets (*/tl/) but not in *s*+sonorant clusters (✓/sl/); and (ii) *s*+sonorant is represented in the same fashion as *s*+obstruent. The implication is that, prior to this, sonority pattern learners like Jarmo do not attend to place identity. Fikkert provides some data consistent with this (e.g., Jarmo's /'drɪŋkə(n)/→['tɪŋkə] 'to drink' (2;4.1)).

Positive evidence for (i) and (ii) comes from medial clusters following stressed vowels: as (22) shows, /tl/, /sl/ and /st/ are syllabified as coda+onset while /tr/ forms a branching onset (Trommelen 1984, Kager & Zonneveld 1985/86).¹⁷ (The initial vowel in (22d) is not underlyingly long which could have prevented heterosyllabification of medial /tr/; it is lengthened because rhymes are minimally bipositional in Dutch.)

(22) Dutch:

- a. ['at'.las] 'atlas'
- b. ['ɔs.lo:] 'Oslo'
- c. ['pas.ta:] 'paste'
- d. ['ma:.tri:ks] 'matrix'

¹⁷ Thanks to Janet Grijzenhout and Stephanie Schreven for help with the transcriptions in (22).

Although data such as these reveal that *s*+sonorant clusters do not pattern as branching onsets, words of the profile in (22a-b) are largely restricted to borrowings and proper names (van der Torre 2003) and are likely infrequent in child-directed speech (Fikkert 1994). Thus, the analysis that *s*+sonorant clusters are not actually branching onsets may be challenging to learn under the scenario that children begin assigning clusters to classes based solely on sonority. Does this mean that adult Dutch is a language where *s*+sonorant clusters are analysed as branching onsets? This is Fikkert's (1994) position but it leaves unexplained the differences in place profile of true branching onsets and *s*+sonorant clusters, as well as the syllabification of *s*+sonorant with *s*+obstruent in word-medial position.

Although (adult) Dutch does not support an analysis where *s*+sonorant clusters are branching onsets, we must question whether there are other languages that may require this analysis. One such language is Sranan, an English-based creole spoken in Suriname. Alber & Plag (2001) report that, in the formation of the creole, English words with true branching onsets and *s*+sonorant clusters were adapted intact, as can be seen in (23a-b). Further, words like *court* and *silver* with post-vocalic liquids commonly underwent metathesis, yielding both obstruent+liquid and *s*+liquid clusters, shown in the last example in each of (23a-b). (23c), however, shows that *s*+stop underwent deletion of *s*. Taken together, the data in (23a-c) suggest that Sranan does not permit appendix *s* and that *s*+sonorant clusters may truly be represented as branching onsets. This analysis, though, raises questions. One concerns the unexpected place profiles of *s*+sonorant if these clusters are truly branching onsets (e.g., [sm] vs. [sn]; see earlier section on cluster phonotactics). Conversely, if *s*+sonorant clusters are instead analysed as appendix/coda-initial, Sranan would be typologically unusual in permitting this cluster type without *s*+stop: all languages with *s*C clusters seem to permit this cluster profile (see section on perceptual considerations below). Clearly, further work must be undertaken on Sranan.

- (23) Sranan:
- a. Obstruent+sonorant:
 promise > pramisi
 trouble > trobi
 court > krutu
 - b. *s*+sonorant:
 smoke > smoko
 snake > sneki
 silver > sriba
 - c. *s*+stop:
 spoil > pori
 stand > tan
 scrape > krebi

We have observed that some researchers consider sonority decisive in assigning clusters to categories with the result that *s*+sonorant clusters are analysed as branching onsets. Consequently, it is *s*+stop that is singled out as formally different. Another approach to the different behaviour of *s*+stop has been to treat these strings as complex segments (Chapter wbctp0029). This view is autosegmentally expressed in (24).

- (24) *s*+stop as complex segment:



Many of the arguments for a single segment analysis of *s*+stop are distributional. Indeed, most researchers who motivate this analysis begin with the observation that all left-edge three-member

clusters in the languages under examination are *s*+stop initial (e.g., Fudge 1969, Selkirk 1982, van de Weijer 1996, Wiese 1996). The position that branching onsets contain a maximum of two consonants can be maintained if *s*+stop form complex segments. As Selkirk (1982) asserts, similar problems can be resolved at the right edge if the single segment analysis holds independent of position (see earlier Fudge 1969). On one hand, words like *next* violate the finding that, inflection aside, English codas are maximally binary. On the other, words such as *wasp* and *task* violate the constraint that *C*₂ in an obstruent coda cluster is always coronal. If these words are segmented as [n-ɛ-k-st], [w-ɑ-sp] and [t-æ-sk], these problems are resolved.

Following on this, Ewen (1982) observes for English that, unlike other clusters which appear in mirror-image order in syllable-initial versus -final position (*clip*, *silk*), *sC* clusters occur in the same order in both positions (*skip*, *task*). He notes that similar observations were made earlier by Vogt (1942) for Norwegian and Sigurd (1965) for Swedish.

A different sort of distributional argument is made by Wiese (1996) for German. He shows that *s*+stop and other complex segments, namely affricates, have similar distributions. Alongside [ʃplit] ‘chippings’ and [ʃpri:sən] ‘to sprout’, we find words like [pfliçt] ‘duty’ and [pfri:m] ‘awl’.

Turning to other types of phonological behaviour, most of the evidence that van de Weijer (1996) forwards involves languages where *s*+sonorant behaves differently from *s*+stop and together with true branching onsets (like Sranan above), thereby supporting a branching onset analysis for *s*+sonorant and another analysis for *s*+stop. Reduplication in Gothic provides the most compelling evidence that the appropriate analysis for *s*+stop is as a complex segment (see also Cairns & Feinstein 1982, Wiese 1996).

Gothic has a class of verbs where the preterite is formed through reduplication. The data in (25a-b) (from van de Weijer 1996) reveal that singleton onsets and the first member of branching onsets are copied. (25c-d) show that *sC* clusters do not behave uniformly. Those with rising sonority pattern with branching onsets –only *C*₁ is copied– while *s*+stop is copied as a unit.

(25) Gothic reduplication:

a.	haita	‘I am called’	hai-hait	‘I was called’
	h ^w opa	‘I boast’	h ^w ai-h ^w op	‘I boasted’
b.	fraisa	‘I try’	fai-frais	‘I tried’
c.	slēpan	‘to sleep’	sai-slēp	‘I slept’
d.	skaida	‘I sever’	skai-skaip	‘I severed’

The pattern in (25d) is as expected if *s*+stop forms a complex segment. Indeed, accounting for it through an appendix/coda+onset representation is challenging at best. Somehow, the analysis would have to copy the consonants up to and including the cluster head.

We turn now to examine some challenges for the complex segment approach to *s*+stop. Although we have provided distributional evidence in favour of this proposal, another look reveals problems on this front. Van der Hulst (1984) rejects the analysis for Dutch on grounds that *s*+stop does not have the same distribution as singleton obstruents. Putative complex segments can follow a vowel but not a consonant, unlike non-complex obstruents: [vɛsp] ‘wasp’ but *[vɛlsp], cf. [vɛlp] ‘lion cub’. The same problem holds for English: [wasp] ‘wasp’, [warp] ‘warp’, *[warsp]. If *s*+stop forms a two-consonant string, this observation follows straightforwardly.

Consider next patterns of epenthesis in cluster repair in L2 acquisition and loanword adaptation (Chapter wbctp0095). (26) presents four patterns of repair for learners/borrowers whose native grammars lack both branching onsets and *sC* clusters (V = epenthetic vowel). Although there are languages where the position of the epenthetic vowel is constant (26a-b), in languages where both prothesis and anaptyxis are observed, two general patterns are found (26c-d): *s*+stop is never interrupted, obstruent+sonorant always is, and *s*+sonorant is variably interrupted.¹⁸

¹⁸ As Fleischhacker (2001) shows, more divisions among *s*+sonorant are actually observed. We return to this below.

(26) Patterns of cluster repair:

a. Japanese (Lovins 1974)	s- <u>V</u> -stop	obstr- <u>V</u> -son	s- <u>V</u> -son
b. Iraqi Arabic (Broselow 1983)	<u>V</u> -s-stop	<u>V</u> -obstr-son	<u>V</u> -s-son
c. Egyptian Arabic (Broselow 1983)	<u>V</u> -s-stop	obstr- <u>V</u> -son	s- <u>V</u> -son
d. Farsi (Karimi 1987)	<u>V</u> -s-stop	obstr- <u>V</u> -son	<u>V</u> -s-son

Broselow (1983) proposes that the reluctance of L2 learners to epenthesize into *s*+stop motivates their analysis as complex segments (see also van de Weijer 1996). While this analysis helps to explain the patterns in (26b-d), it cannot straightforwardly account for the Japanese pattern. Clearly, absence of strings of the shape VsTV (T=stop) in Japanese makes prothesis an impossible repair. Given this, under the view that *s*+stop forms a single segment, the expected pattern for Japanese is simplification of the complex segment, which is van der Weijer's (1996) analysis of Sanskrit reduplication: C₁ is normally copied ([snih]-[si-*snih*] 'to be sticky' (root-perfect), [dru]-[du-druv] 'to run'), but complex segments are reduced to their head (the stop): [skand]-[ka-skánd-a], *[ska-skánd-a] 'to leap'.

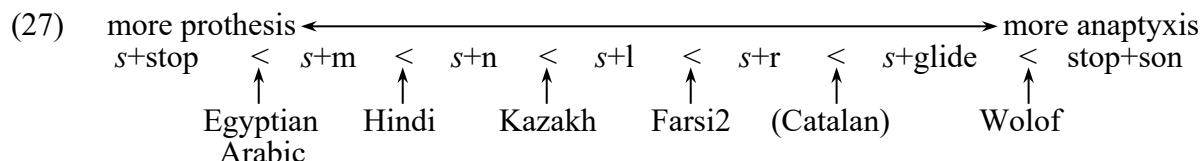
Finally, the single segment hypothesis for *s*+stop is challenged by Byrd (1994) on articulatory grounds. Byrd experimentally examines /sk/ strings in English in three contexts: word-initially, cross-word and word-finally. She observes that #sk involves less overlap than s#k (as well as sk#), and sk# less overlap than gd#. If *s*+stop form single segments, we would expect to find a higher degree of overlap than is observed in strings of consonants that clearly do not form single segments (s#k and gd#).

To sum up, the goal of this section was to examine whether all *s*C clusters pattern as a class, in contrast to branching onsets. On one hand, we observed that *s*+sonorant sometimes patterns with obstruent+sonorant, suggesting that they may have the same representation. Supporting evidence was provided from the developmental path for left-edge clusters in Dutch and possibly from Sranan. On the other hand, we examined evidence in favour of the position that *s*+stop clusters form complex segments. The strongest evidence for this comes from Gothic reduplication. However, both positions were challenged as well, the analysis for child Dutch by the observation that end-state Dutch treats *s*+sonorant together with *s*+obstruent. Perhaps the most damaging evidence against the complex segment analysis is that the articulatory evidence available does not support it.

Perceptual considerations

Throughout the chapter, it has been assumed that a structural difference holds between true branching onsets and (at least some) *s*C clusters. We have observed, however, that languages do not always draw a clear line between these two cluster types: *s*+sonorant sometimes patterns with *s*+stop and at other times with obstruent+sonorant. When we consider Fleischhacker's (2001) survey on epenthesis in L2 and loanword phonology, the problem becomes even more acute: some languages draw the boundary between cluster type *internal* to the *s*+sonorant class, as shown in (27), in contrast to what was observed in (26).¹⁹

¹⁹ Farsi2 refers to the data collected by Fleischhacker. Although Karimi (1987) reports that all *s*+sonorant clusters pattern together, as per (26d), she provides no *s*+rhotic nor *s*+glide examples. The data collected by Fleischhacker show a different pattern for these clusters, as revealed in (27). Since this may reflect a dialect difference, I have labelled the language from which Fleischhacker collected data Farsi2. In addition, Catalan is in parentheses because although it draws a division between *s*+rhotic and *s*+glide, only prothesis is attested; *s*+glide and stop+sonorant do not undergo epenthesis.



Results like these have led Fleischhacker to abandon a structural approach to cluster representation, putting her in the company of others who advocate eliminating the syllable (e.g., Steriade 1999).²⁰ She argues instead for a perceptually-motivated approach to cluster behaviour. Epenthesis site, in particular, is chosen to maximize perceptual similarity between the target (non-epenthesized) form and the output. In view of this, we consider in this section whether the differences that hold between *sC* clusters and true branching onsets can be explained by perceptual considerations alone; this, of course, would challenge the claim that a structural approach to cluster behaviour is necessary.

We do not have space to examine Fleischhacker's proposal in detail but the predictions she motivates are as follows: (i) anaptyxis is preferred to prothesis in *stop+sonorant*; (ii) prothesis is preferred to anaptyxis in *s+stop*; (iii) among *s+sonorant*, more anaptyxis is expected as *C*₂ increases in sonority; and (iv) more anaptyxis is expected in *stop+sonorant* than in *fricative+sonorant*. Concerning (iv), note that Fleischhacker's account does not distinguish *s* from other fricatives; that is, no explanation is provided for the observation that fricatives other than *s* pattern with stops in preferring anaptyxis to prothesis ((26) above). We return to this shortly. First, let us examine the role of perception in *sC* well-formedness in more detail (Chapter wbctp0098).

As alluded to earlier, the acoustic properties of *s*, unlike other obstruents, enable it to appear in positions where it is not followed by a sonorant: strident fricatives have robust internal cues for both place and manner ensuring their perceptibility in all contexts, even before stops (Wright 2004). Clearly, then, the view that segments are ordered to yield a rise in sonority toward the peak does not extend to *s*. Indeed, in spite of the sonority reversal, (strident) fricative+*stop* is superior to both *stop+stop* and *stop+fricative*,²¹ even though the latter two contain a sonority plateau and minimal rise respectively, because (strident) fricatives are less dependent on formant transitions for their identification than stops are (Wright 2004).²²

However, while the acoustic properties of *s* explain why appendices are so often limited to *s* on one hand and why these segments can be followed by stops on the other, they cannot, as far as I can tell, explain cross-linguistic preferences on *sC* profile. (28) shows that *sC* clusters have a rather unusual distribution across languages when viewed from the perspective of perceptual robustness. We focus on word-initial position. Since the perceptibility of all consonants in the *C* position in an initial *sC* cluster will be partly compromised by the preceding *s*, we would expect consonants that are most perceptible to be positioned after *s*. Masking should not be too severe in this context: as mentioned earlier, Byrd (1994) observes that #*sk* clusters involve less overlap than *s#k* and *sk#*. The problem may rather be one of duration: Byrd finds that /*s*/ is longer in #*sk* than in both *s#k* and *sk#*, while /*k*/ is shorter in #*sk* than in both *s#k* and *sk#*. If the relatively short duration of *C* can be generalized to other #*sC* clusters, we would expect segments with robust internal cues to be favoured in this position. Liquids should be optimal since they have clear

²⁰ Steriade's arguments include the absence of clear evidence for word-internal syllable boundaries. Some of the evidence concerns the syllabification of *s+stop* clusters in English when the following vowel is stressed (e.g., *mystérious*). Speakers' judgements vary on where to draw the boundary in such words because neither parse is sanctioned at word edge: *V*₁.st*V*₂ is problematic because *V*₁ is illicit word-finally; *V*₁s.t*V*₂ is problematic because [t] is not aspirated, as it would be word-initially.

²¹ Evidence that this observation is not restricted to *strident* fricative+*stop* comes from Greek: the *stop+stop* clusters in (2a) are often replaced by fricative+*stop*.

²² See Morelli (1999) for an alternative explanation of obstruent cluster well-formedness that appeals to markedness constraints on segment sequencing.

formant structure throughout. Nasals should be favoured over stops since their manner (and to a lesser extent their place) properties are present in the nasal spectrum. Stops, which have weak internal cues, should be the least optimal. In contrast, what we observe in the typology in (28) is that *s*+stop is favoured.²³ In fact, French, Picard and Acoma do not permit *s*+sonorant clusters at all (French has *s*+sonorant in loanwords).

In Goad (2011), the typology provided was limited to Indo-European languages (aside from inclusion of Acoma). As *s*C clusters are sometimes considered to be an Indo-European accident, it is important to show that the typology holds up with inclusion of patterns from non-Indo-European languages. Although a larger sample is required before firm conclusions can be drawn, the table in (28) suggests that *s*+stop > *s*+nasal > *s*+lat > *s*+rhotic (> = is more harmonic than). The favoured profile for *s*C clusters is thus the opposite of that observed for branching onsets. This is not unexpected on a structural account if all *s*C clusters are head-final, in contrast to branching onsets. In *s*C clusters (*s*₁C₂), C₂ is the onset head; thus, it should respect the preferences holding of singleton onsets. Since obstruents are the optimal onsets (e.g., Clements 1990), a parallel should be observed between obstruents in C₁ position in branching onsets (C₁C₂) and stops in C₂ position in *s*C clusters.

(28) *s*C cluster profile across languages:²⁴

<i>s</i> +	Spanish, Basque	French, Picard, Acoma	Greek, Mazateco, Middle Korean	Dutch, Italian, Itonama	Russian, Hungarian, Misantla Totonac
stop	*	✓	✓	✓	✓
nasal	*	*	✓	✓	✓
lateral	*	*	*	✓	✓
rhotic	*	*	*	*	✓

While the asymmetry in the sonority profile of branching onsets versus *s*C clusters follows from the status of *s* as an appendix, it is best captured under Kaye's proposal that *s* is a coda (Goad 2012). Recall from (13) and (22) that medial *s*C clusters in Italian and Dutch are heterosyllabic. If *s*C clusters are *always* coda+onset (Goad 2012, Živanović 2020), then their profile should respect cross-linguistic preferences for syllable contact, which will favour C₂ with lower sonority:

Vs.TV > Vs.NV > Vs.IV > Vs.rV. As C₂ increases in sonority, the cluster prefers to be syllabified as a branching onset, but if this option is never available for *s*C, then higher sonority *s*C clusters will be forbidden, regardless of their position in the word.

The profile in (28) closely parallels Fleischhacker's typology in (27) for preferred epenthesis site in *s*C clusters. Prothesis occurs more commonly when C₂ has lower sonority. As the sonority of C₂ increases, prothesis will result in poor syllable contact. Note as well that the proposed syllable

²³ *s*+fricative has been excluded from the table in (28) because the presence or absence of these clusters in a language does not follow any particular pattern. These clusters present perceptual challenges (Wright 2004) and are generally disfavoured across languages.

²⁴ Notes on individual languages: Recall that *s*C in Acoma might be bogus clusters interrupted by an empty nucleus rather than forming true clusters. In Greek, Middle Korean (Kim-Renaud 1997) and Mazateco (Pike & Pike 1947), *s*+nasal is marginal; note that Steriade (1994) analyses initial clusters in Mazateco as complex segments. Recall that in Dutch, *s*+rhotic may be well-formed in some dialects (note 3); these dialects would fall into the final column with Russian. Itonama is reported to have only *s*+stop and *s*+nasal (Liccardi & Grimes 1968); however, the article includes an example of *s*+lateral: [slahwahn] 'certain' (p. 40; also p. 37), and thus it has been placed in the column with Dutch. For Hungarian, the assessment of which clusters are well-formed comes from Siptár (1980). Finally, Misantla Totonac has no restrictions on *s*C clusters but it contains no *s*+rhotic because it has no /r/ (MacKay 1994), so it has been placed in the column with Russian.

contact account of *s*C well-formedness leads to a distinction between *s*+sonorant and fricative+sonorant, as only the latter can form branching onsets. Thus, the fact that fricative+sonorant patterns with stop+sonorant in epenthesis follows, in contrast to under Fleischhacker's account (see (iv) above).

In sum, I contend that both perceptual and structural considerations must be factored into our understanding of cluster well-formedness. While perceptual considerations can explain why appendices are so often limited to *s* and why *s*+stop is well-formed in spite of its sonority profile, it is the structural differences between *s*C clusters and branching onsets that explain the preference for *s*C profile on the sonority dimension, as well as some observed differences in epenthesis site.

If, as suggested above, *s*C never forms a branching onset, there should be no languages where the only permitted type of *s*C cluster is *s*+liquid (the optimal branching onset according to Clements 1990). This, however, is seemingly violated by Ewe, casting doubt on the typology in (28). In loanwords from English, *s*+stop and *s*+nasal are repaired via epenthesis (Wornyo 2016). *s*+liquid is not, as these clusters, as well as other clusters with liquids in *C*₂, are well-formed in the language; compare (29a) with (29b-e) (data from Stahlke 1971).

- (29) Ewe:
- a. [srá] 'cover'
 - b. [blù] 'growl'
 - c. [gblɔ] 'say'
 - d. [dzra] 'prepare'
 - e. [yru] 'hoot'

Živanović (2020) suggests that all of these clusters are bogus clusters: the two consonants are interrupted by an empty nucleus. I will pursue this analysis here. While the cluster in (29b) looks like an ordinary branching onset, the other data points in (29) suggest otherwise. Under this analysis, (29c-d) would contain branching onsets with complex segments in head position, a cluster type that is disfavoured across languages. (29e) is the most unexpected under a branching onset analysis, as the head of the cluster is a glide, which is higher in sonority than the following liquid. In short, the data in (29) (aside from (29b)) are phonotactically unusual for a branching onset analysis, indeed for any analysis which assumes that the two consonants are truly adjacent.

Conclusion

In this chapter, we have examined several alternative analyses for *s*C clusters. On the empirical front, we have seen that *s*+stop clusters reliably pattern differently from true branching onsets. Not surprisingly, then, the proposals we have examined for *s*+stop all share an important property: *s*+stop clusters are head-final, whether *s* is unaffiliated, an appendix organized by some prosodic constituent above the onset, a coda, or the first member of a complex segment. Branching onsets, by contrast, are head-initial. This difference in headedness helps to explain parallels on the place dimension between *C*₂ in an *s*+stop cluster and *C*₁ in a branching onset, as well as syllabification preferences in word-medial contexts.

Beyond that, however, details of the proposals differ, and following on this, each proposal is both supported and challenged by the available evidence. There are languages like Acoma, English and Dutch where *s* can appear medially after rhymes that are “full”, thereby providing support for the analysis that *s* is linked to the syllable and potentially challenging the coda analysis. However, in some of these same languages, namely English and Dutch, as well as in languages like Italian, the observation that *s*C clusters are heterosyllabic after stressed vowels supports the coda analysis and thereby questions the proposal that *s* is licensed by the syllable or that *s*+stop form a complex segment. The complex segment analysis, in turn, is supported by the reduplication pattern in Gothic which both the appendix and coda analyses fail to elegantly capture.

At present, then, it seems that multiple representations for *s*C clusters may be required. If the number of parametric options is limited and there is robust evidence available for learners to

determine the appropriate representation for the language being acquired, this is far from fatal. For example, the fact that *sC* clusters have a more limited distribution in some languages (Italian) than in others (English, Dutch) can be captured if licensing by the PWd represents the least marked option and therefore the starting point for learners. There will then be positive evidence available to signal learners of some languages that *sC* clusters are licensed lower down, by the syllable. However, we have also seen that this type of scenario may not always work. If Dutch learners initially assume that *s*+sonorant clusters form branching onsets, the evidence available to undo this analysis in favour of one where all *sC* clusters pattern as a class is far from robust and may present a learnability challenge.

The problem with *s*+sonorant more generally is that these clusters are phonotactically ambiguous and, following from this, they pattern ambiguously across languages. While one could fail to be surprised by this, on grounds that these clusters are both *s*-initial and rise in sonority, exactly how their ambiguous behaviour should be formally expressed is far from clear. They appear to be analysed as branching onsets in some languages (e.g., Jarmo's Dutch grammar) and as appendix/ coda-initial in others (e.g., Amahl's English grammar), but in this particular case, this finding is surprising, in view of the otherwise high degree of similarity between the two target languages.

The solution that languages employ different analyses for *s*+sonorant is far from optimal and may lead some researchers to abandon a structural approach to the syllable altogether in favour of a perceptually-grounded account of segmental contact. Indeed, the latter may find support in the observation that even within the class of *s*+sonorant, languages show different patterns of behaviour; we have seen that the division between prothesis and anaptyxis can be drawn anywhere internal to this class. At the same time, however, a purely perceptually-based account seems to be challenged by the finding that preferences for *sC* cluster profile are virtually the inverse of those observed for obstruent+sonorant clusters (Ewe and Sranan aside). While an appeal to syllable contact was made to capture both of these observations, the analysis follows most straightforwardly from the proposal that *sC* clusters are always syllabified as coda+onset strings.

In spite of the quantity of research that had been undertaken on *sC* clusters, it is perhaps most evident that more needs to be done before the issue of their representation can be resolved. A sampling of questions at the two extremes includes the following. At one end of the spectrum, can a more detailed examination of perceptual factors capture differences in the behaviour of fricative+sonorant and *s*+sonorant clusters, thereby further questioning the need for a structurally-based approach to segmental contact and syllabification behaviour? At the other end, if a structural account of behaviour based on syllable contact proves fruitful to pursue, with judicious use of abstract representations, can the coda+onset analysis be motivated for all languages? I leave these and many other questions in between to future research.

References

- Alber, Birgit & Ingo Plag. 2001. Epenthesis, deletion and the emergence of the optimal syllable in creole: The case of Sranan. *Lingua* 11: 811-840.
- Arvaniti, Amalia. 2007. Greek phonetics: The state of the art. *Journal of Greek Linguistics* 8: 97-208.
- Ball, Martin & Nicole Müller. 2016, eds. *Challenging sonority: Cross-linguistic evidence*. Sheffield: Equinox.
- Barlow, Jessica. 1997. *A constraint-based account of syllable onsets: evidence from developing systems*. Bloomington, Indiana: Indiana University dissertation.
- Booij, Geert. 1995. *The phonology of Dutch*. Oxford: Clarendon Press.
- Borowsky, Toni. 1986. Topics in the lexical phonology of English. Amherst, MA: University of Massachusetts dissertation.
- Brockhaus, Wiebke. 1999. The syllable in German: Exploring an alternative. In Harry van der Hulst & Nancy Ritter (eds.), *The syllable: Views and facts*, 169-218. Berlin: Mouton de Gruyter.
- Broselow, Ellen. 1983. Nonobvious transfer: On predicting epenthesis errors. In Susan Gass & Larry Selinker (eds.), *Language transfer in language learning*, 269-280. Rowley, MA: Newbury House.
- Byrd, Dani. 1994. *Articulatory timing in English consonant sequences*. Los Angeles: UCLA dissertation.
- Cairns, Charles & Mark Feinstein. 1982. Markedness and the theory of syllable structure. *Linguistic Inquiry* 13: 193-225.
- Chierchia, Gennaro. 1986. Length, syllabification and the phonological cycle in Italian. *Journal of Italian Linguistics* 8: 5-33.
- Clements, G. Nick. 1990. The role of the sonority cycle in core syllabification. In John Kingston & Mary Beckman (eds.), *Papers in laboratory phonology I: Between the grammar and physics of speech*, 283-333. Cambridge: CUP.
- Cyran, Eugeniusz & Edmund Gussmann. 1999. Consonant clusters and governing relations: Polish initial consonant sequences. In Harry van der Hulst & Nancy Ritter (eds.), *The syllable: Views and facts*, 219-247. Berlin: Mouton de Gruyter.
- Davis, Stuart. 1990. Italian onset structure and the distribution of *il* and *lo*. *Linguistics* 28: 43-55.
- Denzer-King, Ryan. 2009. The distribution of /s/ in Blackfoot: An Optimality Theory account. Missoula, MT: University of Montana MA thesis.
- Derrick, Donald. 2006. Syllabification and Blackfoot “s”. Paper presented at the 38th Algonquian Conference, University of British Columbia, October 27.
- Drachman, Gabarell. 1990. A remark on Greek clusters. In Joan Mascaró & Marina Nespor (eds.), *Grammar in progress: GLOW essays for Henk van Riemsdijk*. Dordrecht: Foris.
- Elsen, Hilke. 1991. *Erstspracherwerb: Der Erwerb des Deutschen Lautsystems*. Wiesbaden: Deutscher Universitäts-Verlag.
- Ewen, Colin. 1982. The internal structure of complex segments. In Harry van der Hulst & Norval Smith (eds.), *The structure of phonological representations*, part 2, 27-67. Dordrecht: Foris.
- Ewen, Colin & Bert Botma. 2009. Against rhymal adjuncts: The syllabic affiliation of English postvocalic consonants. In Phillip Backley & Kuniya Nasukawa (eds.), *Strength relations in phonology*, 221-250. Berlin: Mouton de Gruyter.
- Fikkert, Paula. 1994. *On the acquisition of prosodic structure*. The Hague: HAG.
- Fleischhacker, Heidi. 2002. Cluster-dependent epenthesis asymmetries. In Adam Albright & Taehong Cho (eds.) *UCLA Working Papers in Linguistics* 7: 71-116.
- Fudge, Erik. 1969. Syllables. *Journal of Linguistics* 5: 193-320.
- Fujimura, Osamu & Julie Lovins. 1978. Syllables as concatenative phonetic units. In Alan Bell & Joan Bybee Hooper (eds.), *Syllables and segments*, 107-120. Amsterdam: North-Holland.
- Giegerich, Heinz. 1992. *English phonology*. Cambridge: CUP.
- Gierut, Judith. 1999. Syllable onsets: Clusters and adjuncts in acquisition. *Journal of Speech, Language, and Hearing Research* 42: 708-726.

- Goad, Heather. 2011. The representation of sC clusters. In M. van Oostendorp, C. Ewen, E. Hume & K. Rice (eds) *The Blackwell companion to phonology*, 898-923. Oxford: Wiley-Blackwell.
- Goad, Heather. 2012. sC clusters are (almost always) coda-initial. *The Linguistic Review* 29: 335-373.
- Goad, Heather & Yvan Rose. 2004. Input elaboration, head faithfulness and evidence for representation in the acquisition of left-edge clusters in West Germanic. In René Kager, Joe Pater & Wim Zonneveld (eds.), *Constraints in phonological acquisition*, 109-157. Cambridge: CUP.
- Goad, Heather & Aki Shimada. 2014. /s/ can be a vocoid. In Jyoti Iyer & Leland Kusmer (eds.), *NELS 44: Proceedings of the Forty-Fourth Annual Meeting of the North East Linguistic Society*, 135-148. Amherst, MA: GLSA.
- Goad, Heather & Lydia White. 2006. Ultimate attainment in interlanguage grammars: A prosodic approach. *Second Language Research* 22: 243-268.
- Goldsmith, John. 1990. *Autosegmental and metrical phonology*. Oxford: Blackwell.
- Green, Anthony Dubach. 2003. Extrasyllabic consonants and onset well-formedness. In Caroline Féry & Ruben van de Vijver (eds.), *The syllable in optimality theory*, 238-253. Cambridge: CUP.
- Hall, Tracy Alan. 1992. *Syllable structure and syllable-related processes in German*. Tübingen: Max Niemeyer.
- Harris, James. 1983. *Syllable structure and stress in Spanish*. Cambridge, MA: MIT Press.
- Harris, John. 1994. *English sound structure*. Oxford: Blackwell.
- Hayes, Bruce. 1981. A metrical theory of stress rules. Cambridge, MA: MIT doctoral dissertation.
- Hayes, Bruce. 2009. *Introductory phonology*. Chichester, West Sussex: Wiley-Blackwell.
- Hualde, José Ignacio. 1991. *Basque phonology*. Routledge.
- Hulst, Harry van der. 1984. *Syllable structure and stress in Dutch*. Dordrecht: Foris.
- Iverson, Gregory & Joseph Salmons. 1995. Aspiration and laryngeal representation in Germanic. *Phonology* 12: 369-396.
- Joseph, Brian & Irene Philippaki-Warbuton. 1987. *Modern Greek*. London: Croom Helm.
- Kager, René & Wim Zonneveld. 1985/86. Schwa, syllables, and extrametricality in Dutch. *The Linguistic Review* 5: 197-221.
- Karimi, Simin. 1987. Farsi speakers and the initial consonant cluster in English. In Georgette Ioup & Steven Weinberger (eds.), *Interlanguage phonology: The acquisition of a second language sound system*, 305-318. Cambridge, MA: Newbury House.
- Kaye, Jonathan. 1990. 'Coda' licensing. *Phonology* 7: 301-330.
- Kaye, Jonathan. 1992. Do you believe in magic? The story of s+C sequences. *SOAS Working Papers in Linguistics* 2: 293-313.
- Kaye, Jonathan, Jean Lowenstamm & Jean-Roger Vergnaud. 1990. Constituent structure and government in phonology. *Phonology* 7: 193-231.
- Kenstowicz, Michael. 1994. *Phonology in generative grammar*. Oxford: Blackwell.
- Kim, Chin-Wu. 1970. A theory of aspiration. *Phonetica* 21: 107-116.
- Kim-Renaud, Young-Key. 1997. The phonological analysis reflected in the Korean writing system. In Young-Key Kim-Renaud (ed.), *The Korean alphabet: Its history and structure*, 161-192. De Gruyter Brill.
- Klepousniotou, Katerina. 1998. Onset clusters in Modern Greek. Montréal, Québec: McGill University unpublished manuscript.
- Levin, Juliette. 1985. *A metrical theory of syllabicity*. Cambridge, MA: MIT dissertation.
- Liccardi, Millicent & Joseph Grimes. 1968. Itonama intonation and phonemes. *Linguistics* 38: 36-41.
- Lovins, Julie. 1974. Why loan phonology is natural phonology. In Anthony Bruck, Robert Fox & Michael La Galy (eds.), *Papers from the Parasession on Natural Phonology, Chicago Linguistic Society*, pp. 240-250.

- Lowenstamm, Jean. 1996. CV as the only syllable type. In Jacques Durand & Bernard Laks (eds.), *Current trends in phonology: Models and methods*, 419-443. European Studies Research Institute, University of Salford.
- MacKay, Carolyn J. 1994. A sketch of Misanthla Totonac phonology. *International Journal of American Linguistics* 60: 369-419.
- Miller, Wick. 1965. *Acoma grammar and texts*. Berkeley and Los Angeles: University of California Press.
- Morelli, Frida. 1999. The phonotactics and phonology of obstruent clusters in Optimality Theory. College Park, Maryland: University of Maryland dissertation.
- Myers, Scott. 1987. Vowel shortening in English. *NLLT* 5: 485-518.
- Nespor, Marina & Irene Vogel. 2007. *Prosodic phonology* (2nd ed.). Berlin: Walter de Gruyter.
- Parker, Steve. 2012, ed. *The sonority controversy*. Berlin: De Gruyter.
- Peperkamp, Sharon. 1997. *Prosodic words*. The Hague: Holland Academic Graphics.
- Piggott, Glyne. 1991. Apocope and the licensing of empty-headed syllables. *The Linguistic Review* 8: 287-318.
- Pike, K.L. & E.V. Pike (1947) Immediate constituents of Mazateco syllables. *International Journal of American Linguistics* 13(2): 78-91.
- Polgárdi, Krisztina. 2022. The representation of sC-clusters: A Strict CV analysis. *Glossa* 7(1): 1-29.
- Rice, Keren. 1992. On deriving sonority: A structural account of sonority relationships. *Phonology* 9: 61-99.
- Sebrechts, Koen. 2015. *The sociophonetics and phonology of Dutch r*. Utrecht: LOT.
- Selkirk, Elisabeth. 1982. The syllable. In Harry van der Hulst & Norval Smith (eds.), *The structure of phonological representations*, part 2, 337-383. Dordrecht: Foris.
- Sigurd, Bengt. 1965. *Phonotactic structures in Swedish*. Lund: Uniskol.
- Siptár, Péter. 1980. A note on initial clusters in English and Hungarian. *Acta Linguistica* 30: 327-343.
- Smith, Neil. 1973. *The acquisition of phonology: A case study*. Cambridge: CUP.
- Stahlke, Herbert. 1971. *Topics in Ewe phonology*. Los Angeles, California: UCLA dissertation.
- Steriade, Donca. 1982. Greek prosodies and the nature of syllabification. Cambridge, MA: MIT dissertation.
- Steriade, Donca. 1994. Complex onsets and single segments: The Mazateco pattern. In Charles Kisseberth & Jennifer Cole (eds.), *Perspectives in phonology*, 203-291. Stanford: Center for the Study of Language and Information.
- Steriade, Donca. 1999. Alternatives to syllable-based accounts of consonantal phonotactics. In Osamu Fujimura, Brian Joseph & Bohumil Palek (eds.), *Proceedings of the 1998 Linguistics and Phonetics Conference*, 205-242. Prague: Karolinum.
- Torre, Erik Jan van der. 2003. *Dutch sonorants: The role of place of articulation in phonotactics*. Utrecht: LOT.
- Trommelen, Mieke. 1984. *The syllable in Dutch: with special reference to diminutive formation*. Dordrecht: Foris.
- Tzakosta, Marina. 2009. Asymmetries in /s/ cluster production and their implications for language learning and language teaching. In Anastasios Tsangalidis (ed.) *Selected Papers from the 18th International Symposium of Theoretical and Applied Linguistics*, 365-373. Thessaloniki: Monochromia.
- Tzakosta, Marina & Jeroen Vis. 2009. Phonological representations of consonant sequences: The case of affricates vs. 'true' clusters. In Giorgos K. Giannakis, Mary Baltazani, Giorgos I. Xydopoulos & Tasos Tsaggalidis (eds.) *Proceedings of the 8th International Conference on Greek Linguistics*, 558-573. Department of Linguistics, University of Ioannina.
- Vaux, Bert & Andrew Wolfe. 2009. The appendix. In Eric Raimy & Charles Cairns (eds.), *Contemporary views on architecture and representations in phonology*, 101-143. Cambridge, MA: MIT Press.
- Vennemann, Theo. 1982. Zur Silbenstruktur der deutschen Standardsprache. In T. Vennemann (ed.) *Silben, Segmente, Akzente*, 261-305. Tübingen: Max Niemeyer.

- Vennemann, Theo. 1988. *Preference laws for syllable structure and the explanation of sound change*. Berlin: Mouton de Gruyter.
- Vogt, Hans. 1942. The structure of the Norwegian monosyllable. *Norsk Tidsskrift for Sprogvidenskap* 12: 5-29.
- Waals, Juliette 1999. *An experimental view of the Dutch syllable*. The Hague: HAG.
- Weber, Natalie. 2020. *Syntax, prosody, and metrical structure in Blackfoot*. Vancouver, Canada: University of British Columbia dissertation.
- Weijer, Jeroen van de. 1996. *Segmental structure and complex segments*. Tübingen: Max Niemeyer.
- Wiese, Richard. 1988. *Silbische und Lexikalische Phonologie: Studien zum Chinesischen und Deutschen*. Tübingen: Max Niemeyer.
- Wiese, Richard. 1996. *The phonology of German*. Oxford: Clarendon.
- Wornyo, Albert Agbesi. 2016. English loanwords in Ewe: A phonological analysis. *Journal of Literature, Languages and Linguistics* 22: 42-51.
- Wright, Richard. 2004. A review of perceptual cues and cue robustness. In Bruce Hayes, Robert Kirchner & Donca Steriade (eds.), *Phonetically based phonology*, 34-57. Cambridge: CUP.
- Živanović, Sašo. 2020. Branching onsets 2.0. *Acta Linguistica Academica* 67(1): 73-90.